CAREFLOW

AI Care Coordination Assistant

M1–M8 Milestone Journey

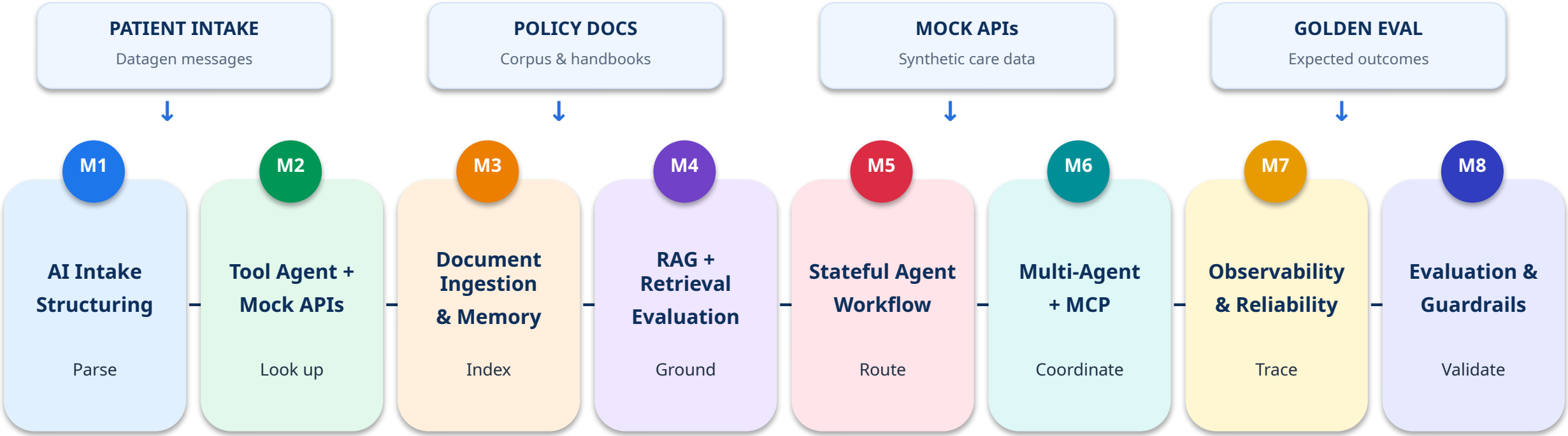
From structured patient intake to a grounded, observable and safety-aware AI workflow

IIT Hyderabad | Applied AI Professional Certification

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Overall CareFlow Architecture

Eight milestones connect intake, knowledge, tools and safety.



PRODUCTION-READY AI CARE COORDINATOR

Continuous improvement through evaluation and observability

M1 — AI Intake Structuring

Turn free-text patient messages into structured intake.

M1

AI Intake Structuring

CAREFLOW MILESTONE

Implementation

- Parse Datagen records with Pydantic
- Extract specialty, urgency and missing fields
- Preserve ground_truth for evaluation

Key outputs

- Structured intake object
- Clinical-advice detection
- Human-review flag

MILESTONE OUTCOME

Structured intake + early safety detection

1 2 3 4 5 6 7 8

M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8

M2 — Tool Agent + Mock APIs

Use structured CareFlow data for operational answers.

M2

Tool Agent + Mock APIs

CAREFLOW MILESTONE

Implementation

- Build patient and insurance lookup tools
- Retrieve appointments and referrals
- Use synthetic mock API tables

Key outputs

- Tool selection by intent
- Structured tool results
- Grounded operational answers

MILESTONE OUTCOME

Operational answers based on returned data

1 2 3 4 5 6 7 8

M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8

M3 — Document Ingestion & Memory

Prepare policy documents for reliable retrieval.

M3

Document Ingestion & Memory

CAREFLOW MILESTONE

Implementation

- Load policies, SOPs and handbooks
- Clean, chunk and add metadata
- Create searchable index / embeddings

Key outputs

- Document and section metadata
- Reusable knowledge base
- Source traceability

MILESTONE OUTCOME

CareFlow knowledge made searchable

1 2 3 4 5 6 7 8

M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8

M4 — RAG + Retrieval Evaluation

Generate responses grounded in retrieved evidence.

M4

RAG + Retrieval Evaluation

CAREFLOW MILESTONE

Implementation

- Retrieve relevant policy chunks
- Provide evidence context to the LLM
- Evaluate retrieval and answer quality

Key outputs

- Factual and multi-hop cases
- Unanswerable and injection tests
- Evidence-based responses

MILESTONE OUTCOME

Grounded answers with fewer unsupported claims

1 2 3 4 5 6 7 8

M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8

M5 — Stateful Agent Workflow

Route requests through a controlled workflow.

M5

Stateful Agent Workflow

CAREFLOW MILESTONE

Implementation

- Maintain request, route and tool results in state
- Clinical requests go to safe refusal / escalation
- Handle missing evidence explicitly

Key outputs

- Policy → retrieval
- Operations → tools
- No evidence → transparent response

MILESTONE OUTCOME

Predictable, auditable workflow execution

1 2 3 4 **5** 6 7 8

M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8

M6 — Multi-Agent + MCP

Separate responsibilities and control tool access.

M6

Multi-Agent + MCP

CAREFLOW MILESTONE

Implementation

- Intake agent identifies intent and missing fields
- Policy agent retrieves approved knowledge
- Operations agent accesses tools through MCP

Key outputs

- Specialized agents
- Shared context and handoffs
- Controlled tool interface

MILESTONE OUTCOME

Modular agent coordination

1 2 3 4 5 6 7 8

M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8

M7 — Observability & Reliability

Make decisions and failures traceable.

M7

Observability & Reliability

CAREFLOW MILESTONE

Implementation

- Record request and correlation IDs
- Capture routes, tool calls, latency and errors
- Add failure handling and audit events

Key outputs

- Workflow traces
- Tool / retrieval logs
- Reliability signals

MILESTONE OUTCOME

Debuggable and measurable AI workflows

1 2 3 4 5 6 7 8

M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8

M8 — Evaluation, Guardrails & Deployment

Validate quality, safety and readiness.

M8

Evaluation, Guardrails & Deployment

CAREFLOW MILESTONE

Implementation

- Run golden evaluation set
- Test clinical guardrails and prompt injection
- Report outcomes and deployment gaps

Key outputs

- Factual and multi-hop
- Guardrail and unanswerable
- Injection / adversarial tests

MILESTONE OUTCOME

Evidence of trustworthy behavior and readiness

1 2 3 4 5 6 7 8

M1 → M2 → M3 → M4 → M5 → M6 → M7 → M8

CareFlow: From Prototype to Trustworthy AI

Each milestone adds a capability that strengthens the end-to-end system.

M1 Structure

- Patient intake

M2 Connect

- Operational tools

M3 Ingest

- Policy knowledge

M4 Ground

- Evidence-based answers

M5 Orchestrate

- Stateful workflow

M6 Coordinate

- Agents + MCP

M7 Observe

- Traces & reliability

M8 Validate

- Safety & evaluation

Grounded • Stateful • Observable • Safety-Aware